



Learning ability is unaffected by isolation rearing in a family-living lizard

Julia L. Riley^{1,2} · Anna Küchler¹ · Théo Damasio¹ · Daniel W. A. Noble² · Richard W. Byrne³ · Martin J. Whiting¹

Received: 15 August 2017 / Revised: 16 December 2017 / Accepted: 22 December 2017 / Published online: 13 January 2018
© Springer-Verlag GmbH Germany, part of Springer Nature 2018

Abstract

The social environment during development can affect learning; for example, raising an obligate social mammal in isolation can hinder their learning ability. However, we know little about how the social environment impacts learning in less-studied, facultatively social taxa, like family-living lizards. We reared tree skinks (*Egernia striolata*) in two treatments, either with a conspecific or in isolation. We used three tasks to quantify skink learning ability (motor, discrimination, and reversal). Skinks performed these tasks under two learning treatments: either after demonstration (social learning) or without social information (individual learning). We did not find any evidence that tree skinks used social information. The majority of skinks learnt our motor (91%) and discrimination tasks (100%), and a third learnt our reversal task (34%). Contrary to our predictions, and the majority of previous literature, we detected no negative effect of rearing treatment on learning in any task. Our surprising findings are likely due to this skink's variable social system, and we suggest that birds and mammals with facultative sociality may not be affected by isolation rearing in the same way as taxa with obligate sociality.

Significance statement

Survival can be impacted by an animal's ability to learn, but many factors can influence this ability (i.e., age, sex, stress, and developmental environment). In this study, we examined how social environment across ontogeny impacts the learning ability of a facultatively family-living animal, the tree skink. Traditionally, the relationship between social environment and learning has been examined in obligate social species. But, examining this relationship in species across all social systems aids in our understanding of the evolution of sociality, and the consequences and limitations of each social system. We found the social environment tree skinks were raised in did not affect their social or individual learning abilities across three foraging tasks. Our findings provide an initial examination of how social environment impacts learning in a facultatively social species.

Keywords Aggregation · Cognition · Facultative sociality · Individual learning · Reptile · Social learning

Communicated by S. J. Downes

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00265-017-2435-9>) contains supplementary material, which is available to authorized users.

✉ Julia L. Riley
julia.riley87@gmail.com

¹ Department of Biological Sciences, Macquarie University, Sydney, NSW, Australia

² Ecology and Evolution Research Centre, School of Biological, Earth, and Environmental Sciences, University of New South Wales, Kensington, NSW, Australia

³ School of Psychology and Neuroscience, University of St. Andrews, St. Andrews, Fife, UK

Introduction

Learning ability is expected to impact survival; for instance, spatial learning ability may benefit foraging, mating opportunities, and predator avoidance, and innovation may enhance survival in novel environments (Dukas 2004). An animal may learn by trial-and-error from information coming directly from the environment (Shettleworth 2010), or, particularly within social species, an individual can utilize social information through the observation of, or interaction with, a conspecific (aka. social learning; Hoppitt and Laland 2013). In both cases, an individual's ability to learn may be affected by a range of factors: for instance, age (Noble et al. 2014; Takahashi et al. 2014), stress level (Boogert et al. 2013; Crino et al. 2014), sex (Einon 1980; Carazo et al. 2014), personality (Sih and Del

Giudice 2012; Trompf and Brown 2014), and developmental environment (i.e., abiotic factors: Clark et al. 2013; Dayananda and Webb 2017; social factors: Thornton and Lukas 2012; Hoppitt and Laland 2013, etc.). Perhaps the most dramatic and best-known factor impacting cognitive development is the social environment during ontogeny.

The effect of early social environment was first demonstrated in 1965, in the impaired learning abilities of socially isolated juvenile rhesus macaques (*Macaca mulatta*; Harlow et al. 1965). Subsequently, many studies have confirmed a negative relationship between isolation rearing and individual learning in social mammals and birds. Yet, this relationship is not always consistent and a variable/positive effect of isolation on learning has also been found (Greenough et al. 1972; Morgan et al. 1975; Einon 1980; Juraska et al. 1984; Wongwitdecha and Marsden 1996; Frisone et al. 2002; Apfelbeck and Raess 2008; Cacioppo and Hawkley 2009; Goerlich et al. 2012). Isolation rearing has the potential to also affect an individual's social learning ability, because it can reduce an individual's ability to associate with conspecifics, and hinder their comprehension of social cues and behavior (e.g., facial signaling, dominance hierarchies; Harlow et al. 1965; Taborsky and Oliveira 2012; Schausberger et al. 2017). This may limit an individual's exposure to social information and/or their ability to use, process, and comprehend social information from conspecifics. Thus, isolation rearing may impact both individual and social learning ability.

To date, the relationship between social developmental environment and learning ability has been predominately examined in obligate social species, limiting its generality. Animal sociality varies from species that are mostly solitary (i.e., forming only temporary aggregations for the purposes of mating) to those that live in complex societies (e.g., eusocial insects) or stable, long-term kin groups (e.g., primates). Restricting the examination of how social developmental environment impacts learning ability to only obligate social species constrains our understanding to only a particular suite of selective forces and does not offer us a comparative evolutionary perspective (Ward and Webster 2016). Animals with facultative sociality are particularly promising systems for research on how social environment impacts development. In species with facultative sociality, parental care is often not required, which allows the social environment to be manipulated unlike in other animals where parental care is obligate (Whiting and While 2017).

Here, we examine the effect of isolation rearing on a facultatively social lizard. Tree skinks (*Egernia striolata*) exhibit facultative, kin-based sociality (Chapple 2003). *Egernia striolata* can be found alone or in groups of variable size (2–10 skinks) mainly consisting of kin (Chapple 2003). Over 80% of aggregations sampled for relatedness consist of parent-offspring relationships or sibling relationships (Duckett et al. 2012). Thus, tree skink social groups are thought to be formed by consecutive litters of offspring

remaining with parents, which results in groups with stable mating pairs and variously aged juveniles (Chapple 2003). Yet, there is intra- and inter-population variation in their social behavior and organization. Aggregation tendency depends on their age, sex, and relatedness (Bonnett 1999). For instance, females are often more solitary than males, and sub-adults are more likely to inhabit crevices with at least two other individuals (Bonnett 1999; Duckett et al. 2012). The social organization of within a tree skink population also varies over time and space, suggesting that seasonal and environmental factors may also influence their social organization (Bustard 1970; Duckett et al. 2012). Interestingly, tree skink social rearing environment also affects their morphological and behavioral development (Riley et al. 2017); a lack of social exposure (e.g., rearing in isolation) and the type of social interactions experienced during development (e.g., competitive or aggressive encounters) both impact juvenile growth and behavior (e.g., sociability, aggression, and boldness; Riley et al. 2017).

Social structure also varies between tree skink populations; for example, within arboreal populations, skinks are mainly found in small groups (maximum of three skinks) or alone (Bustard 1970; Cunningham et al. 2007), but in other arboreal and saxicolous populations, tree skinks form social groups ranging up to a maximum of 10 individuals (Bonnett 1999; Michael and Cunningham 2010; Duckett et al. 2012). The variable nature of tree skink sociality makes it a good model for studying the generality of the relationships between early rearing and learning ability using an experimental framework that manipulates the social rearing environment to mimic what juveniles may experience in the wild. Furthermore, adult female tree skinks socially learn from conspecifics (MJW et al. unpubl. data), allowing us to examine the effects of early rearing on social as well as individual learning ability.

We presented skinks from two rearing treatments (social or isolated) with three learning tasks under two learning treatments (social or individual learning). Our aim was to quantify the impact of social rearing environment on both individual and social learning. We hypothesized that isolation rearing would hinder cognitive development and reduce an individual's likelihood of using social information. We predicted: (1) isolated skinks would be less likely and take longer to learn compared to socially reared skinks, (2) the social learning treatment would be more likely and be faster to learn compared to the individual learning treatment, and (3) that isolated skinks would be less likely to use social information compared to socially reared skinks.

Material and methods

We used 32 tree skinks that were offspring from 19 females collected near Albury, New South Wales (35.98°S, 146.97°E; see [supplementary materials](#) for details about parturition,

husbandry, and measurements). After all juveniles were born, we randomly allocated juveniles into two rearing treatments: either isolated or social.

We used a total of 16 isolated (housed alone), and 16 socially raised (housed within 8 unrelated pairs) juveniles within our experiment. Our captive rearing treatments reflect juvenile social behavior that has been reported in wild tree skink populations; juvenile groups range in size from pairs to 4 individuals, and juveniles have also been observed on their own (Bonnett 1999; Michael and Cunningham 2010; Duckett et al. 2012, JLR unpubl. data). Including parents in the social treatment was not logistically feasible because adult *Egernia* are known to be highly aggressive towards juveniles. Infanticide has been reported in several *Egernia* group spp. (Lanham and Bull 2000; Post 2000; O'Connor and Shine 2004); there are even instances, within captivity, where females eat their own offspring (*E. stokesii*, Lanham and Bull 2000; *E. striolata*, JLR pers. obs. 2015). Thus, due to ethical and logistical considerations, our study was restricted to social groups consisting of juveniles, although wild social groups often consist of parent(s) and offspring (Chapple 2003). Skinks resided within rearing treatments for approximately 1.5 years before we quantified their learning ability (see experimental design below). During development, both isolated and socially reared skinks had limited visual exposure to adult conspecifics during a separate study (details in [supplementary materials](#)).

During the learning trials, we housed juveniles in newspaper-lined, opaque plastic tubs (390 mm W × 580 mm L × 390 mm H) divided in half by a fixed transparent Perspex® divider covered by a removable opaque wooden cover. This allowed us to keep skinks physically separated while also controlling when they viewed each other. The tubs contained a water dish and a refuge (120 mm W × 175 mm L × 38 mm H). A 100-W heat lamp was directed at the refuge, which allowed skinks to thermoregulate and illuminated each tub. We fed skinks three house crickets (*Acheta domesticus*; adults for adult skinks and nymphs for juveniles) dusted with calcium and vitamins once a week after assays were completed. Other than that, the only food the skinks received was the food reward (1.25 ml of puréed fruit; Heinz® apple and pear) that was offered twice daily and eaten only if the learning task was completed successfully.

Learning trial design

Our learning trial was modified from Noble et al. (2014) by altering the food reward from mealworms to 1.25 ml of puréed fruit (Heinz® baby food: apple, and pear). The experiment consisted of motor, discrimination, and reversal learning tasks.

There were two learning treatments: social learning, where skinks observed a conspecific demonstrator correctly executing the task, and individual learning, where skinks simply observed a non-demonstrating conspecific. We allocated an

equal number of isolated and socially reared skinks to each learning treatment (Table S1). We randomly paired an unrelated, adult female skink with each experimental skink for a total of 16 “demonstrators” and 16 “non-demonstrating” conspecifics. The same individuals were paired with each focal skink across all three learning tasks. Demonstrators were trained prior to the experiment with the focal skinks (details in [supplementary materials](#)) and performed tasks correctly in all experimental trials, which ensured that focal skinks received the correct social information.

Prior to trials commencing, we gave skinks 48 h to acclimate to the novel trial housing. At the beginning of all trials, the opaque wooden cover was removed to allow the experimental skink and adult female to view each other's portion of the enclosure through the transparent Perspex® divider. The experimental skink's water dish and refuge were also removed to ensure an unobstructed view. After 20 min of the experimental skink viewing the conspecific, the wooden cover was replaced, the task apparatus was placed within the experimental skink's enclosure, and the experimental skink attempted each task for 1 h. Trials were remotely video-recorded for behavioral scoring. We conducted two trials each weekday, in the morning (0900–1130 h) and the afternoon (1330–1600 h).

Motor task (lid removal)

This task required skinks ($N = 32$; Table 1) to remove an opaque, yellow lid from a dish to gain a food reward (Video S1). Skinks were given 24 trials to complete this task and were classified as learners if they performed 5/6 trials correctly (Table S2). Twenty-nine skinks (91%) learnt this task and moved on to the discrimination task.

Discrimination task

This task required skinks (Table 1) to learn that a blue lid was a reliable cue for a reward, when presented with two dishes on a wooden block (blue vs. white lid; Video S1). Three skinks do not show any significant difference in preference for blue or white (MJW et al. unpubl. data). To control for chemical cues, we placed puréed fruit in both dishes but the reward in the dish with the white lid was inaccessible because of a mesh covering under the lid (Fig. S1). Both lids were removable from both dishes, and this task was scored as being performed correctly if a skink removed the blue lid first (Video S1). A skink could investigate both lids (e.g., via tongue-flicking or visual investigation), but if it removed the white lid first, the task was scored as incorrect. We randomly counter-balanced the blue lid's location across rearing and learning treatments. Skinks were given this task for 24 trials and were classified as learners if they performed 7/8 trials correctly (Table S3). All skinks learnt this task and moved on to the reversal task.

Table 1 Sample sizes (*N*), number of skinks that learnt each task, and the average number of trials it took to learn all tasks (with 95% CIs calculated from raw data) for each rearing and learning treatment

Rearing treatment	Learning treatment	(a) Motor task			(b) Discrimination task			(c) Reversal task		
		<i>N</i>	Number of learners	Number of trials taken to learn	<i>N</i>	Number of learners	Number of trials taken to learn	<i>N</i>	Number of learners	Number of trials taken to learn
Social	Social	8	7	12 (9 to 15)	7	7	10 (7 to 13)	7	1	30
Social	Individual	8	8	13 (9 to 17)	8	8	12 (7 to 16)	8	3	22 (14 to 31)
Isolated	Social	8	7	10 (4 to 15)	7	7	12 (9 to 14)	7	2	24 (14 to 33)
Isolated	Individual	8	7	9 (5 to 13)	7	7	9 (7 to 11)	7	4	28 (17 to 38)

Reversal of discrimination task (hereafter, reversal)

This task was identical to the discrimination task with two exceptions: (1) skinks needed to learn that the white lid now cued reward, instead of the blue lid (Video S1), and (2) skinks were given this task for 34 trials as it was more challenging. Ten skinks (34%) met the reversal task's learning criterion (7/8 trials correct; Table S4).

Behavioral scoring

From video recordings, we scored successful task performance during each trial (either removing the yellow lid, or the correctly colored lid first). Behavioral scorers marked videos blind to skink rearing treatment.

Prior to statistical analysis, we investigated potential inter-observer bias between scorers. JLR and AK scored task success (removing the dish lid) from motor task videos, and JLR, AK, and TD scored task success (removing the correctly colored lid) in discrimination and reversal tasks. Inter-observer reliability statistics were run from data for 21% of the trials for the motor task ($N = 160$), as well as the discrimination and reversal task combined ($N = 145$) using Cohen's Kappa (using the function *cohen.kappa* from the R package *psych* in R v 3.0.3; Kaufman and Rosenthal 2009; R Core Team 2016). For all tasks, inter-observer agreement scores were high (motor task $k = 0.96$; discrimination and reversal task ranged from $k = 0.98$ to 1) and coincided with what is considered "excellent" inter-observer agreement ($k \geq 0.75$; Kaufman and Rosenthal 2009).

Assessment of learning criteria

Prior to statistical analyses, we also needed to assess if our learning criteria were appropriate in categorizing skinks that learnt from those that did not. For this assessment, we used the subset of skinks that had a minimum of six trials after the trial in which they reached the task's learning criterion (Tables S2, S3, S4). Unfortunately, our sample size of skinks that met the learning criterion for the reversal task was too low to perform

the assessment. But for the motor and discrimination tasks, we assessed the robustness of our learning criteria by tallying the number of correct/incorrect choices made by each skink from its final trial needed to reach learning criterion to its very last trial (e.g., if a skink performed 5/6 trials correctly during the motor task, we started the tally at the 6th trial; Noble et al. 2012, 2014; Riley et al. 2016; Tables S2, S3, S4). We tested whether this tally of correct/incorrect choices was significant according to an exact binomial choice test (Carazo et al. 2014; Noble et al. 2014). For the motor and discrimination tasks, respectively, 25/27 (93%) and 25/28 (89%) of skinks performed the task correctly significantly more than expected by chance after meeting the learning criteria. Our findings suggest that our learning criterion for the motor and discrimination tasks was sufficient in categorizing individuals that learnt from those that did not.

Statistical analysis

First, we explored the data to ensure it fitted the assumptions of our analyses (Zuur et al. 2009). During this process, we removed missing values where applicable; thus, there are a variable number of observations and individuals within each model (sample sizes are provided in all result tables; Tables S5, S6, S7).

To fit all our generalized linear mixed effect models (GLMMs), we used the function *glmer* in the *lme4* R package (Bates et al. 2015). For all models, we opted to use the simplest possible model to avoid over-parameterization and issues with model convergence (see details below). The potential confounding factors of skink sex and size were experimentally balanced across rearing and learning treatments to minimize any impact on the data (Table S1). Also, we have previously found that sex does not significantly affect tree skink learning (Riley et al. 2016). For all models, α was set at 0.05. When we predicted fitted lines from the models for visualization, we set the factorial fixed factors to intercept-level values.

Data from motor, discrimination, and reversal tasks were analyzed separately, but the variables included in each of the models were consistent (see Tables S5, S6, and S7 for finalized models):

- (1) This binomial GLMM examined if the *probability of learning a task* (learner = 1, non-learner = 0) was influenced by the *rearing treatment* (isolated or social) or *learning treatment* (social or individual). We also included the additional fixed effect of an interaction between *rearing treatment* $\times$ *learning treatment* but removed this effect and re-ran the model if it was not significant. We accounted for dependency between observations of multiple skinks from each clutch by including a random intercept for *mother identity*. We also included a random intercept for *housing tub* to incorporate dependency among observations of skinks from the same captive environment (e.g., social pairings). We did not analyze the probability of learning the discrimination task because all skinks were successful; this analysis was only performed for the motor and reversal task.
- (2) This Poisson GLMM examined if the *number of trials taken to learn the task* was influenced by the *rearing treatment* (isolated or social) or *learning treatment* (social or individual). If significant, the model also included an interaction between *rearing treatment* $\times$ *learning treatment*. This model also included the random intercepts of *mother identity* and *housing tub*.
- (3) This binomial GLMM examined if *probability of task success during each trial* (either removing the lid or correctly choosing lid color = 1) was influenced by *trial number* (aka. time), *rearing treatment* (isolated or social), or *learning treatment* (social or individual). This model also included the random intercepts of *mother identity* and *housing tub*, as well as a random intercept and slope for *juvenile identity* across trial number to incorporate the dependency among repeated observations of the same individual. If their effect was significant, we also included the additional fixed effects of interactions between *rearing treatment* $\times$ *learning treatment*, *trial number* $\times$ *rearing treatment*, and *trial number* $\times$ *learning treatment*.

We also calculated unconditional means and corresponding 95% CIs (corrected for non-independence) using the function *Effect* in the R package *effect* (Fox 2003; Fox and Hong 2009). Unconditional means of all response variables in each task (*probability to learn*, *number of trials until learnt a task*, and *probability of task success*) were calculated for each rearing and learning treatment. Assessment of unconditional means and the magnitude of their differences (i.e., effects) places more emphasis on biological significance, rather than just statistical significance (which can be affected by sample size) of differences between our treatments (Nakagawa and Cuthill 2007; Gerstner et al. 2017).

Data availability All data and R code from this study can be accessed from the Bitbucket repository at https://bitbucket.org/julia_riley/social-learning-analysis

Results

Motor task

Rearing treatment did not affect whether a skink learnt the motor task ($z = -0.60$, $p = 0.55$; Table 1). Isolated skinks learnt the motor task in, on average, three fewer trials than socially reared skinks ($z = -2.09$, $p = 0.04$; Table 2). The probability of removing the lid during a trial was only an average of 9% higher for isolated skinks, and this difference was non-significant ($z = 1.73$, $p = 0.08$; Table 2).

Learning treatment did not affect whether a skink learnt this task ($z = -0.60$, $p = 0.55$), the number of trials to learn the task ($z = -0.15$, $p = 0.88$; Table 1), or the probability of removing the lid during a trial ($z = 0.86$, $p = 0.39$).

Across trials, the probability of removing the lid increased as trials progressed ($z = 6.95$, $p < 0.01$; Fig. 1A).

Discrimination task

Skinks from all rearing and learning treatments successfully learnt this task. Neither rearing nor learning treatment affected the number of trials necessary to learn the discrimination task (rearing treatment: $z = -0.38$, $p = 0.70$; learning treatment: $z = 0.28$, $p = 0.78$; Table 1), or the probability of making a correct choice (rearing treatment: $z = -1.01$, $p = 0.31$; learning treatment: $z = -0.62$, $p = 0.53$). Across trials, the probability of removing the correct lid increased as trials progressed ($z = 3.67$, $p < 0.01$; Fig. 1B).

Reversal task

Neither rearing nor learning treatment affected whether a skink learnt the task (rearing treatment: $z = 0.84$, $p = 0.40$; learning treatment: $z = 0.16$, $p = 0.88$), the number of trials needed to learn the task (rearing treatment: $z = 0.14$, $p = 0.89$; learning treatment: $z = 0.06$, $p = 0.95$; Table 1), or the probability of making a correct choice (rearing treatment: $z = 0.32$, $p = 0.75$; learning treatment: $z = -0.22$, $p = 0.83$). Across trials, the probability of removing the correct lid increased as trials progressed ($z = 5.29$, $p < 0.01$; Fig. 1C).

Discussion

Our hypothesis that isolation would hinder tree skink cognitive development was not supported; in most cases, we found no effect of rearing treatment on tree skink performance for

Table 2 Unconditional means and 95% CIs for each rearing and learning treatment

	(a) Motor task		(b) Discrimination task		(c) Reversal task	
	Probability of learning the task	Number of trials taken to learn	Probability of trial success	Number of trials taken to learn	Probability of learning the task	Number of trials taken to learn
Rearing treatment						
Social	0.94 (0.67, 0.99)	12.32 (9.84, 15.42)	0.84 (0.68, 0.93)	10.73 (8.95, 12.87)	0.21 (0.03, 0.68)	24.30 (19.69, 29.99)
Isolated	0.88 (0.61, 0.97)	9.18 (7.29, 11.55)	0.93 (0.84, 0.97)	10.24 (8.55, 12.27)	0.41 (0.15, 0.73)	24.80 (20.98, 29.31)
Learning treatment						
Social	0.88 (0.61, 0.97)	10.58 (8.43, 13.28)	0.91 (0.80, 0.96)	10.68 (8.85, 12.88)	0.31 (0.09, 0.69)	24.70 (20.53, 29.72)
Individual	0.94 (0.67, 0.99)	10.79 (8.70, 13.38)	0.87 (0.73, 0.94)	10.32 (8.68, 12.28)	0.28 (0.06, 0.70)	24.50 (20.35, 29.50)

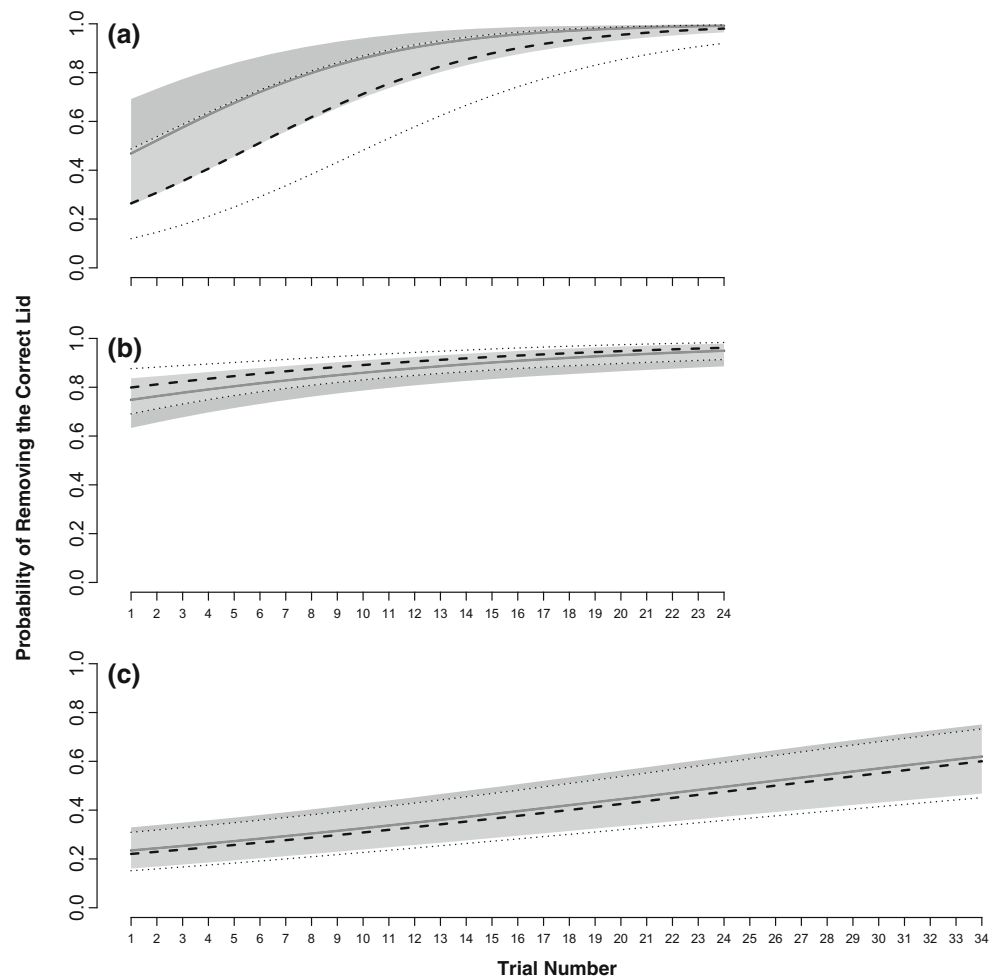
any of our three cognitive tasks. Indeed, isolated skinks learnt the motor task faster than socially reared skinks—a finding opposite to our prediction. However, the effect of rearing treatment on learning speed was weak, as reflected in the conditional means (Table 2). We did not find strong evidence that social environment during tree skink ontogeny affects their learning ability.

Our results contrast with the harmful effects of isolation rearing on cognitive ability that have previously been found in obligate social mammals and birds. Potentially, alternative cognitive tasks may have revealed a negative impact of rearing treatment (Harlow et al. 1965; Thornton and Lukas 2012); but we also failed to find any effect of rearing environment on tree skink individual learning in a previous study, where they were presented with a vertical, spatial maze (Riley et al. 2017). Our findings therefore corroborate previous results on this species and suggest that tree skink learning ability is not negatively affected by developing in social isolation. This finding may be due to the facultative nature of tree skink sociality and parental care (Chapple 2003). Conversely, in mammals and birds where sociality and parental care are largely obligate, the requirements of social exposure for behavioral development may be more fixed. Perhaps a mammal or bird with facultative sociality (e.g., the African striped mouse; Schradin et al. 2011) would show the same, unexpected lack of relationship between isolation and learning ability.

An alternative hypothesis is that the presence or absence of parents, siblings, and/or kin during development may affect tree skink cognition, whereas the presence of an unrelated individual does not. Our rearing treatments did not include parents due to logistical constraints (see above) so these potential effects could not be quantified. Offspring of *Egernia* spp. benefit from the presence of parents by gaining protection, enhanced thermoregulation, and increased access to prey (O'Connor and Shine 2004; Langkilde et al. 2007). Litters of *E. whitii* form size-based dominance hierarchies in which competition reduces growth and increases mortality in the youngest siblings (While and Wapstra 2008); so, the social environment can also be costly for *Egernia* spp. Benefits and costs of living with kin still need to be considered in the cognitive development of *E. striolata*. Furthermore, cognitive development in natural populations may differ from that of our captive-reared individuals; thus, an investigation into the cognitive abilities within natural tree skink populations would be beneficial.

Nonetheless, the social rearing environments within our study were comparable to the social environments found in natural tree skink populations (Riley et al. 2017; JLR unpublished data). Simply because tree skink social pairs were unrelated in this study does not mean that our subjects detected lack of genetic kinship. The mechanism behind kin recognition in *E. striolata* remains unknown and may either be based on phenotypic matching or on cues learnt from

Fig. 1 Probability of removing the correct lid during each trial of the **a** motor, **b** discrimination, and **c** reversal task for isolated skinks (gray 95% CI polygon and solid fitted line) and socially reared skinks (black dotted 95% CI lines and dashed fitted line)



developing in proximity to their natal, family social group (i.e., familiarity; Bull et al. 2001). Although unrelated juveniles were paired within our social rearing treatment, these individuals lacked exposure to related skinks post-birth, so we would expect that individuals would determine their social group based on proximity and familiarity (similar to what may happen via crevice-sharing in the wild; Bull et al. 2001). During development, we observed similar social interactions within our social pairs as have been observed in litters of the closely related *Egernia whitii* (While and Wapstra 2008). Aggressive encounters are common in *E. whitii* litters, which can impact juvenile survival and growth rates (While and Wapstra 2008). We similarly documented antagonistic encounters within social pairs of tree skinks in our study, which increased the incidence of tail loss and affected their growth rates (Riley et al. 2017). Thus, it would be interesting to repeat our study and manipulate relatedness within the social group to investigate if this impacts tree skink cognitive development in a different manner than our findings.

Contrary to our predictions, we also found no evidence that sub-adult tree skinks used social information. Social learning propensity may develop as individuals age (Leris and Reader

2016), or may not be present at particular life stages (Noble et al. 2014). In the context of our study, sub-adults may avoid unfamiliar adults because they can be lethally aggressive (Chapple 2003; O'Connor and Shine 2004). We are therefore hesitant to conclude that tree skinks cannot use social information. Instead, we suggest that sub-adults may not be motivated to use social information from unfamiliar, potentially aggressive, adults. There is prior research that suggests that competition, aggressive interactions, and dominance hierarchies may influence an individual's use of social information (Laland 2004; Kendal et al. 2014; Leris and Reader 2016). For example, young guppies (*Poecilia reticulata*) do not use social information from adults likely because they are not motivated to visit locations where larger adult guppies were present due to the potential for aggression interactions (Leris and Reader 2016). Yet, a guppy's use of social information changed once they reached adulthood, and the size difference between the observer and demonstrator was reduced (Leris and Reader 2016). Future research focusing on social learning between individuals matched in age or between individuals within the same social group (i.e., familiar and/or related) may still find evidence that tree skinks learn socially.

In summary, we did not find that isolation rearing hindered tree skink learning ability. Our study is an initial examination of how social environment impacts individual and social learning in a facultatively social species that is from a taxonomic group (squamate reptiles) underappreciated for their degree of sociality. Our results contrast with the majority of findings in obligate social mammals and birds, and this difference may stem from this species' variable social system and suggests that other facultatively social taxa may also be unaffected by isolation rearing.

Acknowledgments We thank Bram Smagala, Côme Guidou, and Sheila Attersley for field assistance, James Baxter-Gilbert for his assistance with laboratory assays, as well as all the Lizard Lab students that took care of skinks during their development and helped with this research. We also thank two anonymous referees for their helpful comments.

Funding information Financial support for this research was provided by the Australian Research Council (ARC DP130102998 grant to MJW and RWB), Natural Sciences and Engineering Research Council of Canada (scholarship to JLR), the Australasian Society for the Study of Animal Behavior, the Australian Museum, and Macquarie University (scholarship to JLR). DWAN was supported by an ARC Discovery Early Career Research Award (DE150101774), and a University of New South Wales Vice Chancellors Fellowship.

Compliance with ethical standards

Ethical approval Experimental protocols were approved by the Macquarie University Animal Ethics Committee (ARA no. 2013/039). Our research was approved by the New South Wales National Parks and Wildlife Service, Office of Environment and Heritage (License no. SL101264).

Conflict of interest The authors declare that they have no conflict of interest.

References

- Apfelbeck B, Raess M (2008) Behavioural and hormonal effects of social isolation and neophobia in a gregarious bird species, the European starling (*Sturnus vulgaris*). *Anim Behav* 54:435–441
- Bates D, Mächler M, Bolker B, Walker S (2015) Fitting linear mixed-effects models using lme4. *J Stat Softw* 67:1–48
- Bonnett MP (1999) The ecology, behaviour and genetic relationships of a population of *Egernia striolata*. Honours thesis, Flinders University
- Boogert NJ, Zimmer C, Spencer KA (2013) Pre- and post-natal stress have opposing effects on social information use. *Biol Lett* 9(2): 20121088. <https://doi.org/10.1098/rsbl.2012.1088>
- Bull CM, Griffin C, Bonnett M, Gardner MG, Cooper SJB (2001) Discrimination between related and unrelated individuals in the Australian lizard *Egernia striolata*. *Behav Ecol Sociobiol* 50(2): 173–179. <https://doi.org/10.1007/s002650100348>
- Bustard HR (1970) A population study of the scincid lizard *Egernia striolata* in northern New South Wales. *KNAW* 73:186
- Cacioppo JT, Hawkley LC (2009) Perceived social isolation and cognition. *Trends Cogn Sci* 13(10):447–454. <https://doi.org/10.1016/j.tics.2009.06.005>
- Carazo P, Noble DWA, Chandrasoma D, Whiting MJ (2014) Sex and boldness explain individual differences in spatial learning in a lizard. *Proc R Soc B* 281(1782):20133275. <https://doi.org/10.1098/rspb.2013.3275>
- Chapple DG (2003) Ecology, life-history, and behavior in the Australian scincid genus *Egernia*, with comments on the evolution of complex sociality in lizards. *Herpetol Monogr* 17(1):145–180
- Clark BF, Amiel JJ, Shine R, Noble DWA, Whiting MJ (2013) Colour discrimination and associative learning in hatchling lizards incubated at 'hot' and 'cold' temperatures. *Behav Ecol Sociobiol* 68:239–247
- Crino OL, Driscoll SC, Ton R, Breuner CW (2014) Corticosterone exposure during development improves performance on a novel foraging task in zebra finches. *Anim Behav* 91:27–32. <https://doi.org/10.1016/j.anbehav.2014.02.017>
- Cunningham RB, Lindenmayer DB, Crane M, Michael D, MacGregor C (2007) Reptile and arboreal marsupial response to replanted vegetation in agricultural landscapes. *Ecol Appl* 17:609–619
- Dayananda B, Webb JK (2017) Incubation under climate warming affects learning ability and survival in hatchling lizards. *Biol Lett* 13: 2017002
- Duckett PE, Morgan MH, Stow AJ (2012) Tree-dwelling populations of the skink *Egernia striolata* aggregate in groups of close kin. *Copeia* 2012(1):130–134. <https://doi.org/10.1643/CE-10-183>
- Dukas R (2004) Evolutionary biology of animal cognition. *Annu Rev Ecol Syst* 35(1):347–374. <https://doi.org/10.1146/annurev.ecolsys.35.112202.130152>
- Einon D (1980) Spatial memory and response strategies in rats: age, sex and rearing difference in performance. *Q J Exp Psychol* 32(3):473–489. <https://doi.org/10.1080/14640748008401840>
- Fox J (2003) Effect displays in R for generalised linear models. *J Stat Softw* 8:1–9
- Fox J, Hong J (2009) Effect displays in R for multinomial and proportional-odds logit models: extensions to the effects package. *J Stat Softw* 31:1–24
- Frisone DF, Frye CA, Zimmerberg B (2002) Social isolation stress during the third week of life has age-dependent effects on spatial learning in rats. *Behav Brain Res* 128(2):153–160. [https://doi.org/10.1016/S0166-4328\(01\)00315-1](https://doi.org/10.1016/S0166-4328(01)00315-1)
- Gerstner K, Moreno-Mateos D, Gurevitch J, Beckmann M, Kambach S, Jones HP, Seppelt R (2017) Will your paper be used in a meta-analysis? Make the reach of your research broader and longer lasting. *Methods Ecol Evol* 10:236–238
- Goerlich VC, Nätt D, Elfving M, Macdonald B, Jensen P (2012) Transgenerational effects of early experience on behavioral, hormonal and gene expression responses to acute stress in the precocial chicken. *Horm Behav* 61(5):711–718. <https://doi.org/10.1016/j.yhbeh.2012.03.006>
- Greenough WT, Madden TC, Fleischmann TB (1972) Effects of isolation, daily handling, and enriched rearing on maze learning. *Psychon Sci* 27(5):279–280. <https://doi.org/10.3758/BF03328961>
- Harlow HF, Dodsworth RO, Harlow MK (1965) Total social isolation in monkeys. *Proc Natl Acad Sci USA* 54(1):90–97. <https://doi.org/10.1073/pnas.54.1.90>
- Hoppitt W, Laland KN (2013) Social learning: an introduction to mechanisms, methods, and models. Princeton University Press, Princeton. <https://doi.org/10.1515/9781400846504>
- Juraska JM, Henderson C, Müller J (1984) Differential rearing experience, gender, and radial maze performance. *Dev Psychobiol* 17(3): 209–215. <https://doi.org/10.1002/dev.420170302>
- Kaufman AB, Rosenthal R (2009) Can you believe my eyes? The importance of interobserver reliability statistics in observations of animal behaviour. *Anim Behav* 78(6):1487–1491. <https://doi.org/10.1016/j.anbehav.2009.09.014>
- Kendal R, Hopper LM, Whiten A, Brosnan SF, Lambeth SP, Schapiro SJ, Hoppitt W (2014) Chimpanzees copy dominant and knowledgeable individuals: implications for cultural diversity. *Evol Hum Behav* 36: 65–72

- Laland KN (2004) Social learning strategies. *Learn Behav* 32(1):4–14. <https://doi.org/10.3758/BF03196002>
- Langkilde T, O'Connor D, Shine R (2007) Benefits of parental care: do juvenile lizards obtain better-quality habitat by remaining with their parents? *Austral Ecol* 32(8):950–954. <https://doi.org/10.1111/j.1442-9993.2007.01783.x>
- Lanham EJ, Bull CM (2000) Maternal care and infanticide in the Australian skink, *Egernia stokesii*. *Herpetol Rev* 31:151–152
- Leris I, Reader SM (2016) Age and early social environment influence guppy social learning propensities. *Anim Behav* 120:11–19. <https://doi.org/10.1016/j.anbehav.2016.07.012>
- Michael DR, Cunningham RB (2010) The social elite: habitat heterogeneity, complexity and quality in granite inselbergs influence patterns of aggregation in *Egernia striolata* (Lygosominae: Scincidae). *Austral Ecol* 35(8):862–870. <https://doi.org/10.1111/j.1442-9993.2009.02092.x>
- Morgan MJ, Einon DF, Nicholas D (1975) The effects of isolation rearing on behavioural inhibition in the rat. *Q J Exp Psychol* 27(4):615–634. <https://doi.org/10.1080/14640747508400524>
- Nakagawa S, Cuthill IC (2007) Effect size, confidence interval and statistical significance: a practical guide for biologists. *Biol Rev* 82(4):591–605. <https://doi.org/10.1111/j.1469-185X.2007.00027.x>
- Noble DWA, Carazo P, Whiting MJ (2012) Learning outdoors: male lizards show flexible spatial learning under semi-natural conditions. *Biol Lett* 8(6):946–948. <https://doi.org/10.1098/rsbl.2012.0813>
- Noble DWA, Byrne RW, Whiting MJ (2014) Age-dependent social learning in a lizard. *Biol Lett* 10(7):20140430. <https://doi.org/10.1098/rsbl.2014.0430>
- O'Connor DE, Shine R (2004) Parental care protects against infanticide in the lizard *Egernia saxatilis* (Scincidae). *Anim Behav* 68(6):1361–1369. <https://doi.org/10.1016/j.anbehav.2004.02.014>
- Post MJ (2000) The captive husbandry and reproduction of the Hosmer's skink *Egernia hosmeri*. *Herpetofauna* 30:2–6
- R Core Team (2016) R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna <http://www.R-project.org>
- Riley JL, Noble DWA, Byrne RW, Whiting MJ (2016) Does social environment influence learning ability in a family-living lizard? *Anim Cogn* 20:449–458
- Riley JL, Noble DWA, Byrne RW, Whiting MJ (2017) Early social environment influences the behaviour of a family-living lizard. *R Soc Open Sci* 4(5):161082. <https://doi.org/10.1098/rsos.161082>
- Schausberger P, Gratzner M, Strodl MA (2017) Early social isolation impairs development, mate choice and grouping behaviour of predatory mites. *Anim Behav* 127:15–21. <https://doi.org/10.1016/j.anbehav.2017.02.024>
- Schradin C, Lindholm AK, Johannesen J, Schoepf I, Yuen CH, König B, Pillay N (2011) Social flexibility and social evolution in mammals: a case study of the African striped mouse (*Rhabdomys pumilio*). *Mol Ecol* 21:541–553
- Shettleworth SJ (2010) Cognition, evolution, and behavior. Oxford University Press, New York
- Sih A, Del Giudice M (2012) Linking behavioural syndromes and cognition: a behavioural ecology perspective. *Philos Trans R Soc B* 367(1603):2762–2772. <https://doi.org/10.1098/rstb.2012.0216>
- Taborsky B, Oliveira RF (2012) Social competence: an evolutionary approach. *Trends Ecol Evol* 27(12):679–688
- Takahashi K, Masuda R, Yamashita Y (2014) Development of observational learning during school formation in jack mackerel *Trachurus japonicus* juveniles. *Behav Process* 103:52–57. <https://doi.org/10.1016/j.beproc.2013.10.012>
- Thornton A, Lukas D (2012) Individual variation in cognitive performance: developmental and evolutionary perspectives. *Philos Trans R Soc B* 367(1603):2773–2783. <https://doi.org/10.1098/rstb.2012.0214>
- Trompf L, Brown C (2014) Personality affects learning and trade-offs between private and social information in guppies, *Poecilia reticulata*. *Anim Behav* 88:99–106. <https://doi.org/10.1016/j.anbehav.2013.11.022>
- Ward AJW, Webster M (2016) Sociality: the behaviour of group-living animals. Springer International Publishing, Cham
- While GM, Wapstra E (2008) Are there benefits to being born asynchronously: an experimental test in a social lizard. *Behav Ecol* 19(1):208–216. <https://doi.org/10.1093/beheco/arm124>
- Whiting MJ, While GM (2017) Sociality in lizards. In: Rubenstein DR, Abbot P (eds) Comparative social evolution. Cambridge University Press, Cambridge, pp 390–426. <https://doi.org/10.1017/9781107338319.014>
- Wongwitdech N, Marsden CA (1996) Effects of social isolation rearing on learning in the Morris water maze. *Brain Res* 715(1–2):119–124. [https://doi.org/10.1016/0006-8993\(95\)01578-7](https://doi.org/10.1016/0006-8993(95)01578-7)
- Zuur AF, Ieno EN, Elphick CS (2009) A protocol for data exploration to avoid common statistical problems. *Methods Ecol Evol* 1:3–14

Learning ability is unaffected by isolation rearing in a family-living lizard
Behavioural Ecology and Sociobiology
J.L. Riley, A. Küchler, T. Damasio, D.W.A. Noble, R.W. Byrne, M.J. Whiting
Corresponding author: J.L. Riley, Macquarie University, julia.riley87@gmail.com

Supplementary Materials

Material and methods

**Please note that to make the methods as clear as possible, we have repeated some information from the main text.*

Field collection and captive housing

In January 2015, we collected 19 gravid female *E. striolata* from near Albury, New South Wales (35.98°S, 146.97°E). We maintained these skinks at Macquarie University during parturition (from 17 January to 10 February 2016) in a climate-controlled room (maintained at 24°C) within opaque plastic tubs (350 mm W x 487 mm L x 260 mm H). We lined these tubs with newspaper, and placed tree bark, a water dish, and a refuge (120 mm W x 175 mm L x 38 mm H) in each. A UV lamp lighted each skink's tub, and each tub had under-cage heating wire restricted to one side. We cleaned tubs once weekly. We fed skinks 3 adult house crickets (*Acheta domesticus*) dusted with calcium and vitamins twice a week, and puréed fruit (1.25 ml of Heinz® fruit baby food: mango, apple, and pear) once weekly.

During parturition, we checked females twice a day to see if they had given birth. Less than a maximum of 12 h after birth, we measured each juvenile's snout-vent length (mm, SVL: the distance between the tip of the snout and the posterior edge of the cloaca), total length (mm), and tail length (mm) with a clear standard ruler to the nearest 1 mm. We also recorded mass with a digital scale (SP6001, Scout Pro, Ohaus, Pine Brooks NJ, USA) to the nearest 0.01 g, and marked each juvenile with a unique toe-clip (Borges-Landaez and Shine 2003). A total of 32 juveniles were included in our social learning experiment. After they were all born, we randomly allocated juveniles into two rearing treatments: either isolated or social.

Juveniles were housed within smaller opaque plastic tubs than the females (85 mm W x 140 mm L x 60 mm H), either alone if in the isolated treatment or within their social pair. All other housing criteria were the same as the adults', except they were fed nymph *A. domesticus* instead of adults. During the winter (June to August), we maintained skinks at a reduced temperature (18°C). Across their development, all individuals from both treatments were visually exposed to adult conspecifics while quantifying their sociability in a separate study (Riley et al. 2017). This sociability assay

exposed each juvenile to four unrelated, adult, female conspecifics during four behavioural trait assays that were 5 h in length, and which occurred at ca. 5, 7, 9, and 12 months of age (Riley et al. 2017). During this assay, the juveniles could not smell or touch the adult female because they were separated via a fixed transparent Perspex[®] divider (Riley et al. 2017). Thus all skinks within this study were habituated to visually encountering a variety of adult tree skinks prior to their exposure to them in this study. Three days prior to this experiment, on 12 August 2016, we re-measured each skink's morphometrics and recorded their sexes.

Training of Demonstrators

The demonstrator lizards (N = 16) were trained on each task during the week prior to each of the three tasks being given to the experimental, focal lizards (e.g., each week before the motor, discrimination, and reversal tasks). In the case of the discrimination and reversal task (i.e., the two dish task with blue and white lids), we fixed the lid to the incorrect dish on the demonstrator's apparatus so that it was not removable. This ensured the experimental, focal lizards received reliable information (Noble et al. 2014).

REFERENCES

- Borges-Landaez PA, Shine R (2003) Influence of toe-clipping on running speed in *Eulamprus quoyii*, an Australian scincid lizard. *J Herp* 37:592–595.
- Noble DWA, Byrne RW, Whiting MJ (2014) Age-dependent social learning in a lizard. *Biol Letters* 10:20140430–20140430. doi: 10.1098/rsbl.2014.0430
- Riley JL, Noble DWA, Byrne RW, Whiting MJ (2017) Early social environment influences the behaviour of a family-living lizard. *R Soc Open Sci* 4:161082. doi: 10.1098/rsos.161082

VIDEOS

Video S1. *E. striolata* attempting the (a) motor, (b) discrimination and (c) reversal task. On the bottom left are skinks that are attempting the task after viewing a demonstrator performing the same task correctly (social learning treatment), and on the bottom right are skinks attempting the task after simply viewing an adult female (individual learning treatment). Both skinks perform the motor (removing the yellow lid off the dish to access food), and the discrimination task (accessing food from under the blue lid) correctly. Only the skink on the left performs the reversal task correctly (accessing food from under the white lid). Videos are sped up 4x, and edited to highlight successful task performance.

TABLES

Table S1. Each experimental lizard's (N = 32) rearing treatment (I = isolated, S = social), learning treatment (SL = social, IL = individual), the identity of their mother (PIT-tag number), housing tub number (a shared alphanumeric code indicates social pairs), sex (M = male, F = female), snout-vent-length (SVL, mm), and mass (g).

Skink	Rearing Treatment	Learning Treatment	Mother Identity	Housing Tub	Sex	SVL (mm)	Mass (g)
B1122	I	IL	1468494	S19	F	102	22.7
B1121	I	IL	1468537	S28	F	94	19.0
B1110	I	IL	1468538	S29	M	95	21.5
B1022	I	IL	1468544	S30	M	91	20.7
B1011	I	IL	1469229	S33	F	91	16.7
B0323	I	IL	1468506	S22	F	95	20.5
B1010	I	IL	1468549	S31	M	93	21.2
B0230	I	IL	1468460	S16	F	97	21.9
B0212	I	SL	1468504	S21	F	99	24.5
B0220	I	SL	1468529	S26	M	100	22.7
B0302	I	SL	1468480	S17	F	95	20.7
B0322	I	SL	1468515	S24	F	96	22.3
B0331	I	SL	1468532	S27	M	95	21.9
B0333	I	SL	1468532	S25	F	98	23.7
B1000	I	SL	1468509	S23	F	98	22.7
B1031	I	SL	1468510	S32	F	97	21.7
B1012	S	IL	1468476	P24	F	76	10.8
B0303	S	IL	1468506	P16	M	90	19.6
B0320	S	IL	1468515	P18	F	92	17.9
B0330	S	IL	1468532	P19	F	76	10.3
B0332	S	IL	1468516	P18	F	92	20.1
B0213	S	IL	1468529	P19	F	99	24.1
B0214	S	IL	1468529	P20	M	96	23.8
B0223	S	IL	1468460	P13	F	90	15.7
B0200	S	SL	1468515	P17	M	90	19.1
B0203	S	SL	1468506	P12	F	90	18.9
B0221	S	SL	1468460	P11	F	92	19.7
B0222	S	SL	1468460	P12	M	90	18.8
B1001	S	SL	1468509	P17	F	99	20.6
B1002	S	SL	1468549	P21	F	98	22.7
B1030	S	SL	1468491	P25	F	88	17.5
B1032	S	SL	1468502	P25	F	93	20.6

Table S2. Tally of correct (1) and incorrect (0) task performance for our motor task. The learning criterion (5/6 correct choices) is outlined for each skink. The trial at which each skink met our learning criterion for the motor task is bolded and italicized. The trials that we used to assess robustness of our learning criterion are shaded in grey. Skink rearing treatment (I = isolated, S = social), learning treatment (SL = social, IL = individual), number of trials taken to meet learning criterion, each skink's learning categorization (learner = Y, non-learner = N), tally of correct/incorrect trials for the assessment of learning criterion, and binomial probability of each assessment of learning criterion are also specified.

Skink	Rearing Treatment	Learning Treatment	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	T19	T20	T21	T22	T23	T24	Number of Trials until Learnt	Learner	Tally of Correct Choices	Binomial Probability
B0200	S	SL	1	0	1	0	1	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	12	Y	13/13	<0.001
B1122	I	IL	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	6	Y	18/19	<0.001	
B0203	S	SL	1	1	0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	14	Y	11/11	<0.001	
B1121	I	IL	1	1	1	0	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	11	Y	14/14	<0.001	
B0212	I	SL	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	Y	19/19	<0.001	
B1110	I	IL	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	Y	19/19	<0.001	
B0220	I	SL	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	Y	19/19	<0.001	
B1022	I	IL	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	Y	18/19	<0.001	
B0221	S	SL	0	1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	10	Y	15/15	<0.001	
B1012	S	IL	0	0	0	0	0	1	0	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	14	Y	10/11	0.005	
B0222	S	SL	0	0	0	0	0	0	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	15	Y	10/10	0.001	
B1011	I	IL	0	0	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	1	1	1	0	1	0	19	Y	4/6	0.234	
B0302	I	SL	0	0	1	0	1	0	0	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	N			
B0303	S	IL	1	0	1	0	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	8	Y	16/17	<0.001	
B0322	I	SL	0	0	1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	11	Y	14/14	<0.001	
B0320	S	IL	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	7	Y	17/18	<0.001	
B0331	I	SL	1	1	1	0	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	11	Y	14/14	<0.001	
B0323	I	IL	0	0	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	8	Y	15/17	0.001	
B0333	I	SL	1	1	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	6	Y	18/19	<0.001	
B0330	S	IL	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	13	Y	11/12	0.003	
B1000	I	SL	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	Y	19/19	<0.001	
B0332	S	IL	0	0	0	1	0	1	0	1	1	0	0	0	1	1	1	1	1	1	1	1	0	1	0	18	Y	4/7	0.273	
B1001	S	SL	0	0	0	0	0	0	1	1	1	0	1	0	0	1	0	0	0	1	0	1	1	1	0	N				
B1010	I	IL	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	7	Y	18/18	<0.001	
B1002	S	SL	0	0	1	0	1	1	1	0	1	1	1	1	1	1	0	0	0	1	0	1	1	1	1	10	Y	11/15	0.042	
B0213	S	IL	0	1	0	0	0	0	0	1	1	1	1	0	1	0	1	1	1	1	1	1	1	1	1	13	Y	11/12	0.003	
B1030	S	SL	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	Y	19/19	<0.001	
B0214	S	IL	0	1	0	0	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	10	Y	14/15	<0.001	
B1031	I	SL	0	0	0	0	1	0	1	0	0	1	0	0	0	0	1	0	1	0	1	1	1	1	0	22	Y			
B0223	S	IL	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	1	1	1	1	1	20	Y			
B1032	S	SL	1	0	1	0	1	1	1	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	17	Y	8/8	0.004	
B0230	I	IL	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	1	0	0	1	1	N				

Table S3. Tally of correct (1) and incorrect (0) choices for our discrimination task. The learning criterion (7/8 correct choices) is outlined for each skink. The trial at which each skink met our learning criterion for the discrimination task is bolded and italicized. The trials that we used to assess robustness of our learning criterion are shaded in grey. Skink rearing treatment (I = isolated, S = social), learning treatment (SL = social, IL = individual), number of trials taken to meet learning criterion, each skink's learning categorization (learner = Y, non-learner = N), tally of correct/incorrect trials for the assessment of learning criterion, and binomial probability of each assessment of learning criterion are also specified.

Skink	Rearing Treatment	Learning Treatment	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	T19	T20	T21	T22	T23	T24	Number of Trials until Learnt	Learner	Tally of Correct Choices	Binomial Probability
B0200	S	SL	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	8	Y	16/17	<0.001
B1122	I	IL	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	8	Y	17/17	<0.001	
B0203	S	SL	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	9	Y	15/16	<0.001	
B1121	I	IL	0	0	0	1	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	14	Y	11/11	<0.001	
B0212	I	SL	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	9	Y	15/16	<0.001	
B1110	I	IL	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	8	Y	16/17	<0.001	
B0220	I	SL	0	1	1	1	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	14	Y	11/11	<0.001	
B1022	I	IL	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	8	Y	16/17	<0.001	
B0221	S	SL	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	8	Y	16/17	<0.001	
B1012	S	IL	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	9	Y	16/16	<0.001	
B0222	S	SL	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	0	1	8	Y	15/17	0.001	
B1011	I	IL	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	8	Y	15/17	0.001	
B0303	S	IL	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	8	Y	16/17	<0.001	
B0322	I	SL	1	1	0	1	1	0	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	15	Y	9/10	0.010	
B0320	S	IL	1	1	1	0	1	1	0	1	0	1	1	1	0	0	1	0	1	1	1	1	0	1	1	24	Y			
B0331	I	SL	1	1	1	0	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	13	Y	11/12	0.003	
B0323	I	IL	0	1	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	1	0	1	0	9	Y	11/16	0.067	
B0333	I	SL	0	1	1	0	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	0	1	1	1	14	Y	9/11	0.027	
B0330	S	IL	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	9	Y	15/16	<0.001	
B1000	I	SL	0	0	1	1	1	1	1	1	1	0	1	1	0	1	1	1	0	1	1	1	1	1	1	10	Y	12/15	0.014	
B0332	S	IL	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	9	Y	16/16	<0.001	
B1010	I	IL	1	1	1	1	1	0	1	1	1	0	1	0	1	0	1	1	0	0	1	0	1	1	0	1	8	Y	10/17	0.148
B1002	S	SL	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	8	Y	17/17	<0.001	
B0213	S	IL	1	1	1	0	1	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	14	Y	11/11	<0.001	
B1030	S	SL	0	1	1	1	0	0	1	0	1	0	1	0	1	1	1	1	1	0	1	0	1	1	1	17	Y	6/8	0.109	
B0214	S	IL	1	1	1	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	0	1	1	0	12	Y	11/13	0.010	
B1031	I	SL	1	1	1	1	1	1	0	1	0	0	1	0	0	1	1	1	1	0	1	1	1	1	1	8	Y	12/17	0.047	
B0223	S	IL	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	9	Y	16/16	<0.001	
B1032	S	SL	0	1	1	1	0	1	1	1	0	1	1	1	1	1	0	1	1	0	1	1	1	1	1	13	Y	10/12	0.016	

Table S4. Tally of correct (1) and incorrect (0) choices for our reversal task. The learning criterion (7/8 or 8/10 correct choices) is outlined for each skink. The trial at which each skink met learning criterion for the reversal task is bolded and italicized. The trials that we used to assess robustness of our learning criterion are shaded in grey. Skink rearing treatment (I = isolated, S = social), learning treatment (SL = social, IL = individual), number of trials taken to meet learning criterion, each skink's learning categorization (learner = Y, non-learner = N), tally of correct/incorrect trials for the assessment of learning criterion, and binomial probability of each assessment of learning criterion are also specified.

Skink	Rearing Treatment	Learning Treatment	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	T19	T20	T21	T22	T23	T24	T25	T26	T27	T28	T29	T30	T31	T32	T33	T34	Number of Trials until Learnt	Learner	Tally of Correct Choices	Binomial Probability	
B0200	S	SL	0	0	0	1	0	0	1	0	0	0	0	0	0	1	1	1	1	0	1	1	0	0	1	1	1	1	0	0	1	1	0	0	0	0	33	Y			
B1122	I	IL	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	1	0	0	0	1	1	1	0	0	1	1	1	1	1	2	1				
B0203	S	SL	1	0	1	1	0	0	1	0	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	1	0	0				
B1121	I	IL	0	1	0	0	1	0	1	1	1	0	0	1	0	1	0	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	0	0	0	0	1				
B0212	I	SL	0	1	0	0	1	1	0	1	1	0	1	0	1	1	1	0	1	1	1	2	0	0	0	0	1	1	0	1	1	0	1	1	1	1	0	20	Y	9/15	0.153
B1110	I	IL	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	1	1	0	1	1	1	0	0	1	0	0	1				
B0220	I	SL	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0	0	0	1	0	0	0	1	1	1					
B1022	I	IL	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	1	0	1	0	1	1	0	0					
B0221	S	SL	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	1	1	0	1	1	0	0	0	1					
B1012	S	IL	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
B0222	S	SL	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	1	0	0	1	0	0	0	0					
B1011	I	IL	0	0	0	1	1	0	0	1	0	0	0	0	0	1	0	1	1	1	0	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0				
B0303	S	IL	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1	1	0	0	0	0	0	0	0	1	0	1	1	1	0	1	1	1				
B0322	I	SL	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	1	1	1	1	0	1	2	1	1	0	0	0	0	1	0	0	1	1	1	0	22	Y	6/13	0.209
B0320	S	IL	0	0	0	0	1	0	1	0	1	0	0	0	0	1	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1			
B0331	I	SL	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	1	1	1	1	2	0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	20	Y	11/15	0.042
B0323	I	IL	0	0	1	0	0	1	0	1	0	1	0	0	0	1	1	1	0	0	0	0	1	2	0	1	1	1	1	1	1	1	1	1	1	1	1	22	Y	12/13	0.002
B0333	I	SL	1	0	1	0	0	0	0	1	1	0	0	0	0	1	1	0	0	1	0	0	1	1	0	0	1	1	1	0	1	1	1	1	2	1	0	32	Y		
B0330	S	IL	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	1	1	0	1	2	1	1	1	0	1	1	1	1	1	0	0	0	1	22	Y	10/13	0.035
B1000	I	SL	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	1	0	0	1	0	0	1	1	1	0	1	1	1	0	0	0	0	1	1	0				
B0332	S	IL	1	0	0	0	0	1	1	0	0	0	0	0	0	1	0	1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1				
B1010	I	IL	0	1	0	0	0	1	1	0	0	0	1	1	0	0	1	0	1	0	1	1	1	1	1	0	0	0	1	0	0	1	0	1	0	1	1	1			
B1002	S	SL	0	0	1	0	0	0	0	0	0	1	1	0	0	1	1	0	0	1	0	0	1	0	1	1	1	0	0	0	0	0	1	0	1	1	0				
B0213	S	IL	0	0	0	0	0	1	1	1	1	0	0	0	1	0	0	0	1	0	1	0	1	0	0	1	1	1	1	0	1	1	1	1	1	1	1	26	Y	9/9	0.002
B1030	S	SL	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	1	1	1	1	0	1	1	0	1	1	1	1	1	1	1	30	Y		
B0214	S	IL	0	1	0	0	0	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	1	0	0	1	0	0	1	1	0	0	1	1	0	0	0				
B1031	I	SL	0	1	0	0	0	0	0	0	0	1	0	1	0	1	0	0	1	0	1	1	0	1	1	0	1	1	0	1	1	0	1	1	1	0	1				
B0223	S	IL	0	1	1	1	0	0	0	0	1	0	0	0	1	1	1	1	1	1	2	1	1	0	0	0	1	1	1	1	0	0	0	1	1	1	0	19	Y	11/16	0.067
B1032	S	SL	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	0	0	1	0	1	0	0	0	1	0	1	1	0				

Table S5. Outcomes of mixed effect models that examined the effect of a skink's rearing treatment (ISOLATED or SOCIAL), learning treatment (social learning; SL or individual learning; IL), and the interaction between our two treatments on (1) the probability of learning the motor task, (2) number of trials until learnt the motor task, and (3) probability of removing the lid during each trial. Our third model also included the additional fixed factors of trial number, and interactions between trial number x rearing and learning treatment; if a fixed factor was not included in a model it is indicated with 'na'. If models initially included a fixed factor, but it was removed from the final full model due to non-significance it is indicated with '—'. Parameter estimates for binomial models are on the *log odds* link scale, and Poisson models are on the *log link* scale. Significance is indicated using an asterix (*).

<i>Fixed Effects</i>	Probability of Learning <i>N_{ind} = 32, N_{mom} = 19, N_{tub} = 27</i>				Number of Trials until Learnt <i>N_{ind} = 29, N_{mom} = 19, N_{tub} = 27</i>				Probability of Removing the Lid <i>N_{obs} = 768, N_{ind} = 32, N_{mom} = 19, N_{tub} = 27</i>			
	β	SE	z-value	P	β	SE	z-value	P	β	SE	z-value	P
Intercept (SOCIAL, IL, F)	3.158	1.343	2.351	0.019 *	2.521	0.125	20.117	<0.001 *	-1.239	0.506	-2.451	0.014 *
Trial Number	na	na	na	na	na	na	na	na	0.215	0.031	6.954	<0.001 *
Rearing Treatment (ISOLATED)	-0.771	1.287	-0.599	0.549	-0.295	0.141	-2.094	0.036 *	0.898	0.518	1.733	0.083
Learning Treatment (SL)	-0.771	1.287	-0.599	0.549	-0.020	0.130	-0.153	0.879	0.449	0.519	0.864	0.387
Rearing Treatment * Learning Treatment	—	—	—	—	—	—	—	—	—	—	—	—
Trial Number * Rearing Treatment	na	na	na	na	na	na	na	na	—	—	—	—
Trial Number * Learning Treatment	na	na	na	na	na	na	na	na	—	—	—	—

Table S6. Outcomes of mixed effect models that examined the effect of a skink's rearing treatment (ISOLATED or SOCIAL), learning treatment (social learning; SL or individual learning; IL), and the interaction between our two treatments on (1) number of trials until learnt the discrimination task and (2) probability of a correct choice during each trial. Our second also included the additional fixed factors of trial number, and interactions between trial number x rearing and learning treatment; if a fixed factor was not included in a model it is indicated with 'na'. If models initially included a fixed factor, but it was removed from the final full model due to non-significance it is indicated with '—'. Parameter estimates for binomial models are on the *log odds* link scale, and Poisson models are on the *log* link scale. Significance is indicated using an asterix (*).

<i>Fixed effects</i>	Number of Trials until Learnt <i>N_{ind}</i> = 29, <i>N_{mom}</i> = 19, <i>N_{tub}</i> = 25				Probability of Making Correct Choice <i>N_{obs}</i> = 696, <i>N_{ind}</i> = 29, <i>N_{mom}</i> = 19, <i>N_{tub}</i> = 25			
	β	SE	<i>z-value</i>	<i>P</i>	β	SE	<i>z-value</i>	<i>P</i>
Intercept (SOCIAL, IL)	2.357	0.105	22.530	<0.001 *	1.300	0.305	4.264	<0.001 *
Trial Number	na	na	na	na	0.080	0.022	3.672	<0.001 *
Rearing Treatment (ISOLATED)	-0.047	0.122	-0.382	0.702	-0.290	0.286	-1.013	0.311
Learning Treatment (SL)	0.034	0.121	0.277	0.781	-0.177	0.284	-0.623	0.534
Rearing Treatment * Learning Treatment	—	—	—	—	—	—	—	—
Trial Number * Rearing Treatment	na	na	na	na	—	—	—	—
Trial Number * Learning Treatment	na	na	na	na	—	—	—	—

Table S7. Outcomes of mixed effect models that examined the effect of a skink's rearing treatment (ISOLATED or SOCIAL), learning treatment (social learning; SL or individual learning; IL), and the interaction between our two treatments on 1) the probability of learning the reversal task, (2) number of trials until learnt the reversal task, and (3) probability of a correct choice during each trial. Our third model also included the additional fixed factors of trial number, and interactions between trial number x rearing and learning treatment; if a fixed factor was not included in a model it is indicated with 'na'. If models initially included a fixed factor, but it was removed from the final full model due to non-significance it is indicated with '—'. Parameter estimates for binomial models are on the *log odds* link scale, and Poisson models are on the *log* link scale. Significance is indicated using an asterix (*).

	Probability of being a Learner <i>N_{ind}</i> = 29, <i>N_{mom}</i> = 19, <i>N_{tub}</i> = 25				Number of Trials until Learnt <i>N_{ind}</i> = 29, <i>N_{mom}</i> = 19, <i>N_{tub}</i> = 25				Probability of Making Correct Choice <i>N_{obs}</i> = 986, <i>N_{ind}</i> = 29, <i>N_{mom}</i> = 19, <i>N_{tub}</i> = 25			
<i>Fixed effects</i>	β	<i>SE</i>	<i>z-value</i>	<i>P</i>	β	<i>SE</i>	<i>z-value</i>	<i>P</i>	β	<i>SE</i>	<i>z-value</i>	<i>P</i>
Intercept (SOCIAL, IL)	-1.391	1.210	-1.149	0.250	3.186	0.107	29.658	<0.001 *	-1.313	0.238	-5.512	<0.001 *
Trial Number	na	na	na	na	na	na	na	na	0.051	0.010	5.294	<0.001 *
Rearing Treatment (ISOLATED)	0.944	1.130	0.835	0.404	0.020	0.143	0.143	0.887	0.081	0.255	0.317	0.751
Learning Treatment (SL)	0.158	1.009	0.157	0.876	0.008	0.140	0.058	0.954	-0.055	0.254	-0.218	0.827
Rearing Treatment * Learning Treatment	—	—	—	—	—	—	—	—	—	—	—	—
Trial Number * Rearing Treatment	na	na	na	na	na	na	na	na	—	—	—	—
Trial Number * Learning Treatment	na	na	na	na	na	na	na	na	—	—	—	—

FIGURES



Figure S1. The set-up for our discrimination and reversal task: two dishes were placed on a wooden block. One dish had a white lid, and the other a blue lid. During the discrimination task the food reward was under the blue lid, and during the reversal task the food reward was under the white lid. Under the incorrect lid, the food was made inaccessible by plastic mesh.